DreamVLA

From Generated Human Motions to Humanoid Vision-Language-Action Policies

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ember-center-berkeley.github.io/DreamVLA

Generated motion → residual expert → vision-language-action policy

01 Generate demonstrations

Replace repeated task-specific teleoperation with generated motion and physically executed expert demonstrations.

1 Generate and retarget
Task text + contact constraints → OmniControl human motion → OmniRetarget G1 references.

2 Adapt contact with residual RL
Preserve frozen SONIC tracking priors with a privileged residual expert that tracks both robot and task objects under randomized physics.

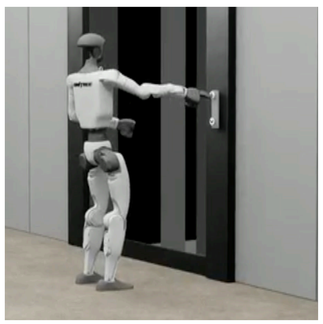
3 Keep successful expert rollouts
Record egocentric RGB, proprioception, executed tokens, hand commands, and language in LeRobot format. Initialize GROOT N1.7 on successful episodes.

Executed token = SONIC token + learned residual

Four simulated G1 interactions


Pick from platform


Open drawer


Press button


Walk and pick

02 Recover from learner errors

Offline demonstrations cover expert states. Learner errors alter contacts and move the VLA away from that distribution.

1 Roll out the current VLA
Execute images + proprioception + language. Save simulator states along the learner trajectory.

↓

2 Restore a failed learner state
Sample a time uniformly from the failed rollout. Restore that state and resume the same reference suffix.

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3 Collect an expert continuation
The privileged expert attempts recovery. Keep the corrective trajectory only when it completes the task.

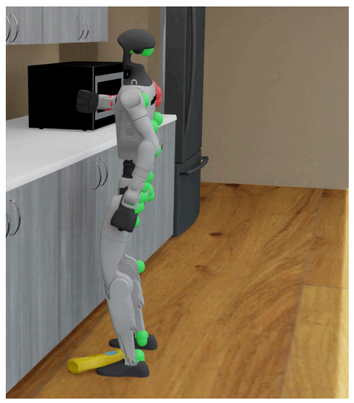
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4 Aggregate and fine-tune
Add successful reference-conditioned continuations, build action chunks within each segment, and repeat for three DAgger rounds.

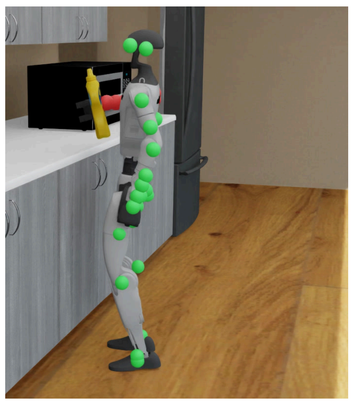
Student deployment
RGB + robot state + instruction → token / hand actions → frozen SONIC decoder. No privileged object state or reference input.

Residual adaptation at contact

Nominal SONIC

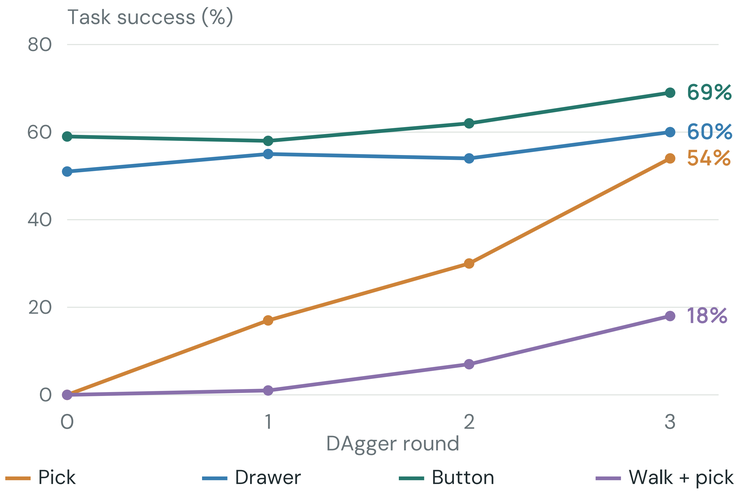


Residual SONIC



Nominal tracking loses the bottle; residual execution retains it.

03 Measure closed-loop success



100 random environment settings per task; paper Table 2.

Corrective data vs. more offline data

Task	Offline	DAgger-3	Offline control
Pick	0%	**54%**	0%
Drawer	51%	**60%**	54%
Button	59%	**69%**	57%
Walk + pick	0%	**18%**	0%

200 initial + 200 successful corrections per round = 800 total trajectories. Offline control: 800 expert-only trajectories.

Generated-data scale: offline only

Refs / episodes	Pick	Drawer	Button	Walk + pick
10 / 10	0%	13%	0%	0%
50 / 50	0%	51%	14%	0%
200 / 200	0%	51%	59%	0%

Main finding
Synthetic data can yield both task success and policy robustness with 54% pick, 60% drawer, 69% button, and 18% walk-and-pick success at 800 trajectories, eliminating the need for expensive humanoid teleoperation.